VU Medical Zone

CHE 301 Final Term Current Fall 2020

Qno1: What type of energy occur when energy absorb by ion or molecule?

There are various **types** of electromagnetic radiation, there are various **types**. Depending on the **energy** of the photon **absorbed** or emitted, different phenomena can **occur**. When an **atom absorbs** an UV photon or a photon of visible light, the **energy** of that, Hence, the color observed is of the metal **ion** only.

Qno2: What happened when condensed ring increase?

As the number of condensed rings increases, the absorption is shifted progressively to longer wavelength.

Qno3: Differentiate between partition coefficient and distribution coefficient?

The success of this method depends upon the difference in solubility of a compound in various solvents. For a given compound solubility differences between solvents is quantified as the "distribution coefficient"

When a compound is shaken in a separatory funnel with two immiscible solvents, the compound will distribute itself between the two solvents. It is known as partition coefficient.

Qno4: Write three application of IR spectroscopy.

Applications of IR Spectroscopy

1. Structure Determination

Search for benzene ring presence. Search for functional groupregions (carbonyl, NH, OH, CO etc.) • Search for CH of different types.

2. Identification of compound

• Matching given structure (metal, complex formation)

• Matching of standard through finger print region

• Differentiate if we have alcohol or ether

3. Progress of reaction

• Oxidation of secondary alcohol to ketone

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• Reduction of alcohol to alkene

• Hydrolysis of sugars

Qno5: Write down the type or source used in IR spectroscopy.

Radiation Source:

• Nernest filament is composed of a mixture of oxides of zirconium, thorium and yttrium, which is formed into a hollow rod about 2 mm in diameter and 2-5 cm in length.

• Globar is a bonded silicon carbide rod about 7 mm in diameter and 5 cm in length.

Qno6: What is flash chromatography.

Flash Chromatography:

In flash column chromatography the sample is pumped through the packed glass column. Flash silica has quite smaller mesh size.

Qno7: Explain the function of ICP.

ICP (Inductively Coupled **Plasma**) Spectroscopy is an analytical method used to detect and measure elements to analyze chemical samples. The process is based on the ionization of a sample by extremely hot **plasma**, usually made from argon gas.

Qno8: What are the types of Sorbents?

Polar Sorbent:

POLAR INTERACTION

• Also called hydrophilic or normal phase

• Involves hydrogen bonding, pi-pi and dipole/ dipole interactions

• Sorbents - silica, diol, diethylamino, cyanopropyl

• Applications - lipids, oil additives, carbohydrates, phenols, oil soluble vitamins.

Non-Polar Sorbent:

• It is also called reverse phased.

• Where polar silica's ends are replaced with nonpolar carbon chain.

• Same as in HPLC.

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Ion exchange sorbent:

• Cation exchange resins are negatively charged and adsorb positively charged cations from matrix.

• Anion exchange resins are positively charged and adsorb negatively charged conjugate bases from matrix.

Copolymeric sorbent:

Hydrophobic & ionic retention mechanisms. Reverse phase sorbent with cation or anion exchange.

Acidic, basic & neutral analyte applications.

Matrix aqueous.

Elution solvents mixture of organics with acid or bas.

Qno9: Explain the chiral chromatography.

CHIRAL CHROMATOGRAPHY

Special chemically modified stationary phases, for example cellulose or amylose derivatives, proteins or peptides, cyclodextrins etc., for the separation of enantiomers

Most separations are based upon partition mechanisms utilizing chemically modified silica as the stationary phase and polar solvents as the mobile phase. The surface of the support, for example the silanol groups of silica, is reacted with various silane reagents to produce covalently bound silyl derivatives covering a varying number of active sites on the surface of the support. The nature of the bonded phase is an important parameter for determining the separation properties of the chromatographic system.

Qno10: Name of the detectors used in Spectroscopy.

1. Photomultiplier tubes

2. UV/Vis Spectrophotometers

3. Phototube

4. Diode array detector

Qno11: Dimensions of column.

Classification of Column Dimensions:

Standard column: Diameter 4.6 mm, length 15 to 25 cm

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Micro Bore Columns (MBC): Columns with internal diameters of less than 2 mm are often referred to as MBC. Mobile phase consumption 1/20 compared to that in 4.6 mm diameter. Small injection volume required.

Small Bore Column (SBC): Columns with internal diameter around 2.5 mm referred to as SBC. Mobile phase consumption 1/3 compared to that in 4.6 mm diameter.

Fast-LC columns: These columns packed with 3µm dia, column packing is just 5 cm long permit very fast analysis.

Qn12: Working of ICP.

- The sample is prepared in liquid form as in case of FAAS.
- Sample is injected through nebulizer @ 0.5 to 1.5 mL/min.
- The emitted spectral lines pass through, monochromator.
- The separation in MS is on,basis of mass to charge ratio.

Qno13: What is SPE?

SPE is a useful technique used to extract out the minute concentrations of analyte from our matrix. Solid-phase extraction (SPE) is a sample preparation process by which compounds that are dissolved or suspended in a liquid mixture are separated from other compounds in the mixture according to their physical and chemical properties

Qno14: Explain the Precipitation gravimetry in detail.

Precipitation gravimetry, the analyte is separated from a solution of the sample as a precipitate and is converted to a compound of known composition that can be weighed.

Precipitation Gravimetry

- The analyte is converted to a sparingly soluble precipitate which are then filtered, washed (free of impurities), converted to a product of known composition by suitable heat treatment, and weighed.
- For example, a precipitation method for determining calcium in natural waters. The reactions are:

2NH 3 + H2C2O4 C2O4-2	2NH4+ +

Ca+2

(aq) + C2O4-2 (aq)

CaC2O4 (s)

Precipitation Gravimetry:

- The precipitates are filtered using a weighed filtering crucible, then dried and ignited to converts the precipitate to calcium oxide. The reaction is:

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$CaC_2O_4 (s) \Delta CaO (s) + CO (g) + CO_2$

• After cooling, the crucible and precipitate are weighed, and the mass of calcium oxide is determined by subtracting the known mass of the crucible. The calcium content of the sample is then computed.

Qno15: Write down the characteristics of an absorption band.

The two main characteristics of an absorption band are;

Its position: Corresponds to wavelength of radiation whose energy is equal to that required for the electronic transition.

Its intensities: Corresponds to the number of molecules which absorb the radiation of that wavelength, depends largely on probability of transition. It may be measured as Transmittance (T) or Absorbance (A).

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